

ZERO-QUERY DEPTH-ONE QAOA SURFACES FROM LOCAL EDGE STATISTICS

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Abstract. Depth-one MaxCut QAOA admits an exact classical surface formula determined by edge degrees and common-neighbor counts. This paper implements the formula as a streamed zero-query baseline and evaluates it before learned singular-value priors. Edge statistics require $O(n + m + \sum_{(u,v)} \min\{\deg u, \deg v\})$ preprocessing; a q -point surface then costs $O(mq)$ arithmetic and $O(1)$ extra storage when streamed. The implementation is checked against independent statevectors at all 3,888 values from 48 authenticated ten-vertex neighborhoods of Stanford Gset G1. Maximum absolute discrepancy is 1.33×10^{-15} . Group-isomorphism analysis shows that these 48 tasks form only 16 classes and that 38 of 48 assignments in a superseded row split have an isomorphic training task. Class-grouped folds remove that leakage. At one queried target value, mean normalized regret is 0.01572 for a training mean, 0.00920 for ridge, and 0.00791 for a singular-value prior; the exact formula has zero regret at every budget while using no target objective queries. The result replaces a query-efficiency claim by an exact computational baseline and a collision-aware evaluation contract. It applies to unweighted depth one, not higher depth or hardware execution.

1. Depth-one objective and source tasks. For an unweighted graph $G = (V, E)$, define

$$C = \sum_{(u,v) \in E} \frac{I - Z_u Z_v}{2}, \quad B = \sum_{v \in V} X_v, \quad (1.1)$$

and

$$F_G(\gamma, \beta) = |E|^{-1} \langle + | e^{i\gamma C} e^{i\beta B} C e^{-i\beta B} e^{-i\gamma C} | + \rangle. \quad (1.2)$$

The registered grid contains nine values of each angle. Forty-eight tasks are deterministic ten-vertex breadth-first neighborhoods of the checksum-pinned Stanford Gset G1 graph [2]. They are related samples from one graph, not independent graph draws.

Theorem 1 (Isomorphism invariance). *If G' is obtained by a vertex permutation, then $F_{G'}(\gamma, \beta) = F_G(\gamma, \beta)$ for every angle pair and every QAOA depth. Isomorphic tasks cannot be separated across an independence-based train-test split.*

Proof. Let U_π permute qubits. Then $C_{G'} = U_\pi C_G U_\pi^\dagger$, $U_\pi B U_\pi^\dagger = B$, and $U_\pi |+\rangle^{\otimes n} = |+\rangle^{\otimes n}$. Conjugating every layer and the measured cost in Eq. (1.2) by U_π leaves the scalar expectation unchanged. $\square$

Exact isomorphism testing gives class sizes 11, 7, 7, 5, 4, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1. Four grouped folds contain 12 tasks each. The old row folds place an isomorphic training task beside 38 of 48 test assignments; Fig. 1 reports the full overlap audit.

2. Exact streamed surface. For edge (u, v) , set $d_u = \deg(u) - 1$, $d_v = \deg(v) - 1$, and $\lambda_{uv} = |N(u) \cap N(v)|$.

Theorem 2 (Exact depth-one edge value). *At depth one,*

$$\begin{aligned} \langle C_{uv} \rangle &= \frac{1}{2} + \frac{1}{4} \sin(4\beta) \sin \gamma (\cos^{d_u} \gamma + \cos^{d_v} \gamma) \\ &\quad - \frac{1}{4} \sin^2(2\beta) \cos^{d_u+d_v-2\lambda_{uv}} \gamma [1 - \cos^{\lambda_{uv}}(2\gamma)]. \end{aligned} \quad (2.1)$$

Summing Eq. (2.1) and dividing by $m = |E|$ gives the exact objective. With hashed adjacency, preprocessing costs $O(n + m + \sum_{(u,v)} \min\{\deg u, \deg v\})$ and $O(n + m)$ storage. A q -point surface costs $O(mq)$ work and may be streamed in $O(1)$ extra storage.

Proof. Mixer conjugation gives

$$e^{i\beta B} Z_u Z_v e^{-i\beta B} = c^2 Z_u Z_v + cs(Z_u Y_v + Y_u Z_v) + s^2 Y_u Y_v, \quad (2.2)$$

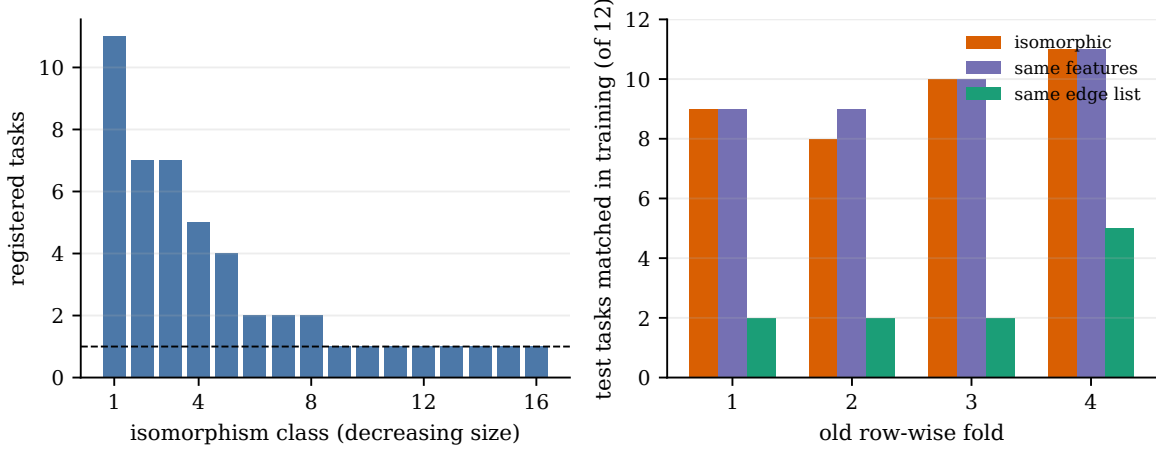


Fig. 1: Isomorphism-class multiplicities and leakage counts in the superseded row-wise split. Grouped folds assign each class wholly to one test fold.

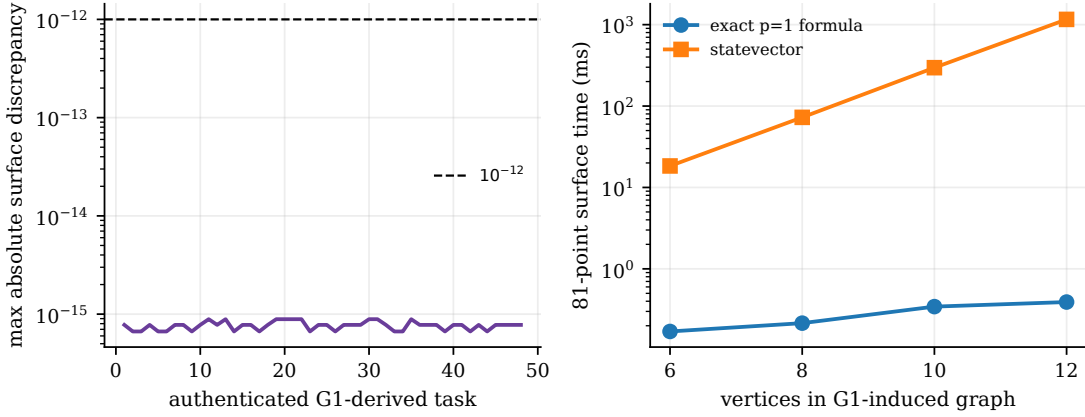


Fig. 2: Independent statevector verification and matched full-surface CPU time. All 3,888 values agree near unit roundoff; the formula avoids exponential state storage.

with $c = \cos(2\beta)$ and $s = \sin(2\beta)$. Cost terms outside the closed neighborhood of (u, v) commute and cancel. Conjugating the remaining Pauli strings and using $\langle +|P|+ \rangle = 0$ unless $P \in \{I, X\}^{\otimes n}$ leaves the two endpoint contributions and one common-neighbor contribution in Eq. (2.1); this is the established Wang–Hadfield–Jiang–Rieffel identity [1]. Degrees follow from one adjacency pass, while intersecting the smaller endpoint-neighbor set costs $O(\min\{\deg u, \deg v\})$ per edge. Formula evaluation is constant work per edge and angle pair. $\square$

Figure 2 tests the exact expression against independent statevector evaluation and reports the matched runtime scaling.

3. Grouped query audit. Each test surface is ranked by a predicted surface. Regret after querying the first b ranked target values is

$$r_b = \frac{\max_{\theta} F(\theta) - \max_{\theta \in Q_b} F(\theta)}{\max_{\theta} F(\theta)}. \quad (3.1)$$

This finite-grid statistic does not assert continuous global optimality. Table 1 gives the low-budget grouped-fold values before the full curves are resolved graphically.

Figure 3 shows the complete budget curve and fold variability. The singular-value residual passes its registered five-percent sampled-row gate in only two of four folds and does not consistently beat ridge. The exact formula returns the entire target surface classically and therefore has zero regret before any target objective evaluation.

Table 1: Grouped-fold mean normalized regret. The exact formula uses zero target objective queries; learned methods use the stated budget.

Method	$b = 1$	$b = 2$	$b = 4$
Training mean	.015721	.014183	.003326
Ridge	.009196	.003326	.000000
Spectral prior	.007912	.007849	.006823
Exact formula	.000000	.000000	.000000

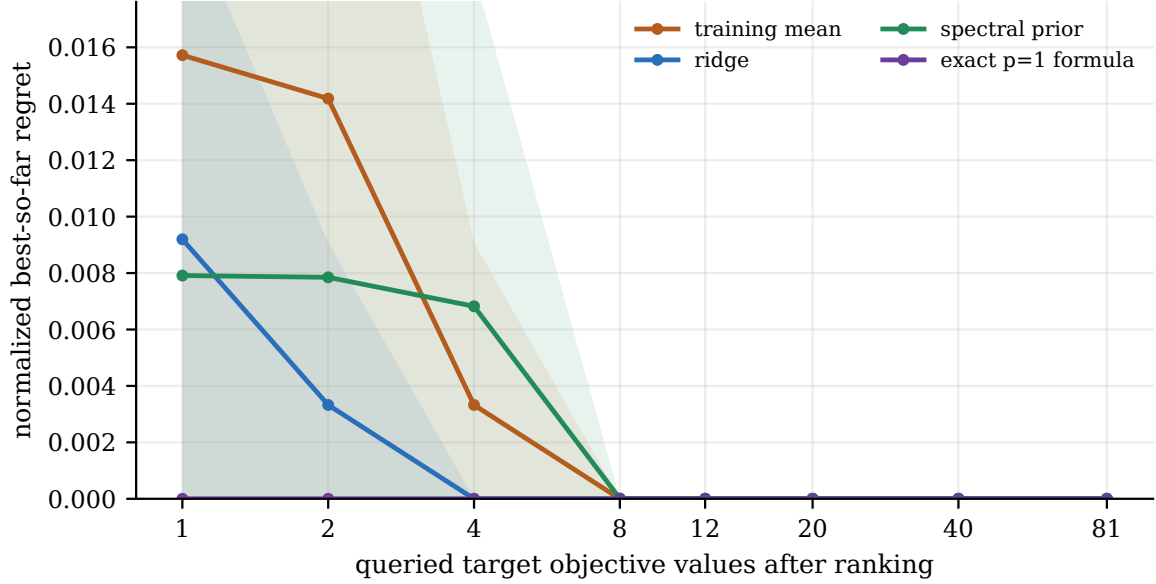


Fig. 3: Collision-aware grouped-fold regret versus target objective queries. Shading is the fold standard deviation.

4. Scope. The exact baseline is specific to unweighted depth-one MaxCut and a fixed angle grid. It does not extend unchanged to higher depth, weighted objectives, noise, or hardware sampling. The authenticated tasks come from one Gset graph. Within that scope, every learned query-advantage claim is dominated by the zero-query formula, and row-wise evaluation is invalidated by graph collisions. The useful output is therefore the streamed formula implementation, its resource bound, and an isomorphism-grouped testing protocol.

References.

- [1] Zhihui Wang, Stuart Hadfield, Zhang Jiang, and Eleanor G. Rieffel. Quantum approximate optimization algorithm for MaxCut: A fermionic view. *Physical Review A*, 97:022304, 2018. doi: 10.1103/PhysRevA.97.022304. Corrected 30 November 2020.
- [2] Yinyu Ye. Gset max-cut benchmark instances. Stanford University, n.d. URL <https://web.stanford.edu/~yye/yye/Gset/>. Accessed 31 July 2026.